

2 Further calculus	2.1	Further Integration — Reduction formulae.	Students should be able to derive formulae such as: $nI_n = (n-1)I_{n-2}, n \geq 2, \text{ for}$ $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx,$ $I_{n+2} = \frac{2 \sin(n+1)x}{n+1} + I_n$ for $I_n = \left(\int \frac{\sin nx}{\sin x} dx \right) [\text{CP1}]$
	2.2	The calculation of arc length and the area of a surface of revolution.	The equation of the curve may be given in Cartesian, parametric or polar form.